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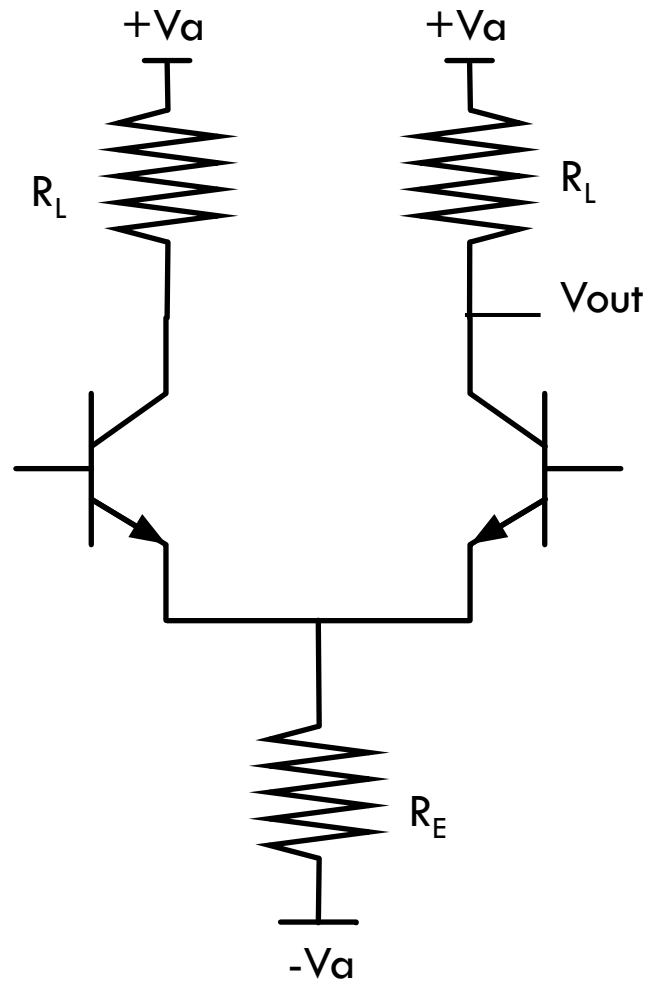
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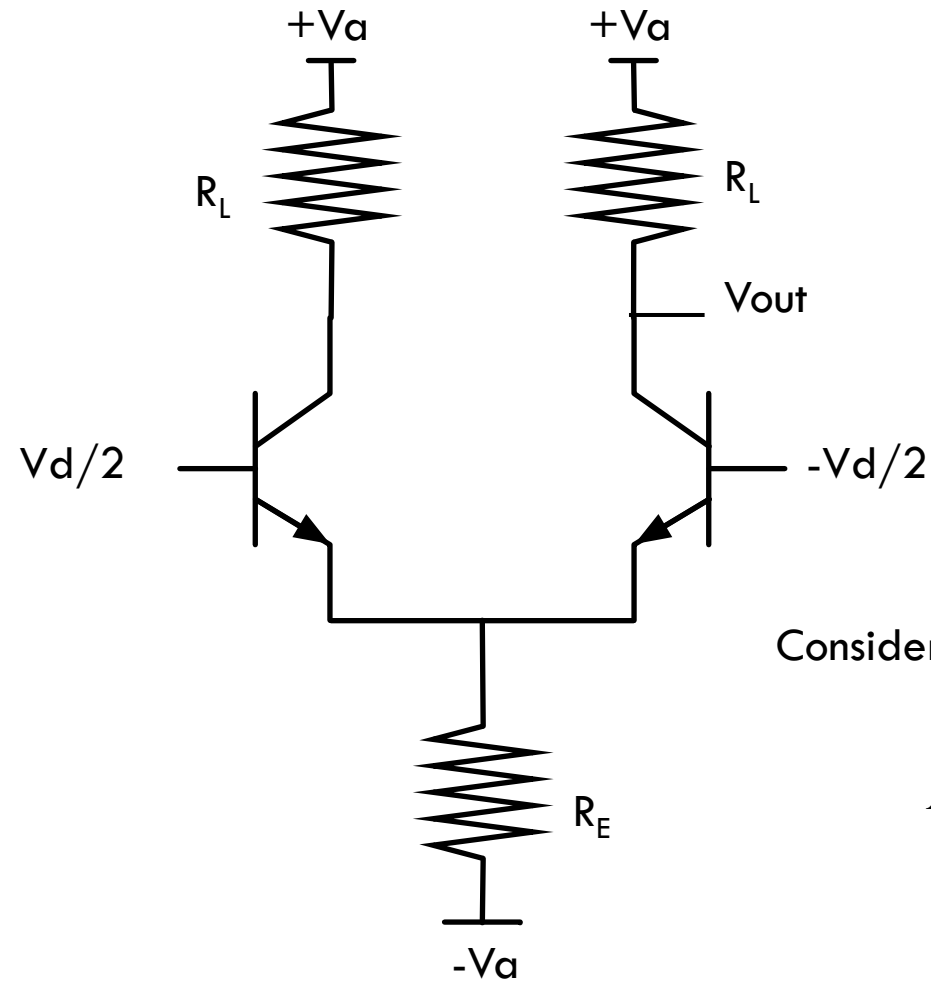
BASIC ANALOG STAGES

Associate professor Stefano Sagginì
Lezione di circuiti e sistemi elettronici

Stadi differenziali



Stadi differenziali

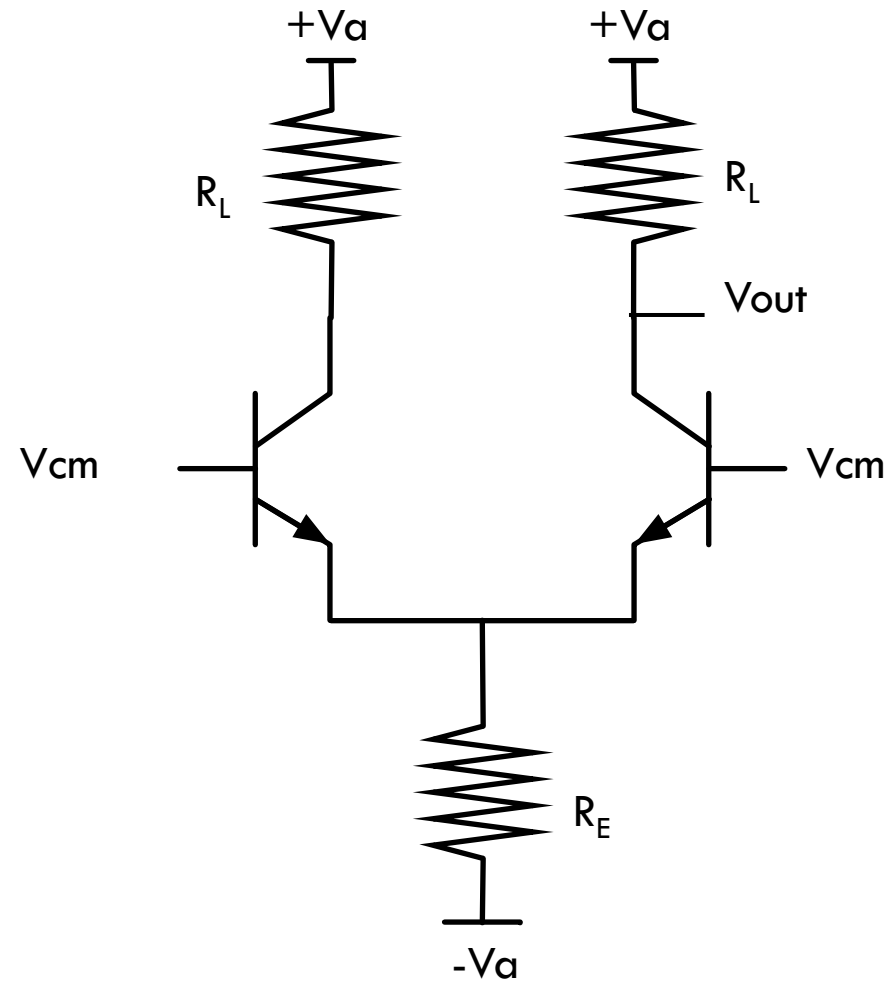


$$A_d = \frac{gm}{2} R_L = \frac{IR_L}{2V_T}$$

Considerando $IR_L = V_a$

$$A_d = \frac{V_a}{2V_T}$$

Stadi differenziali



$$A_{cm} = -\frac{R_L}{2R_E + 1/gm} \approx -\frac{R_L}{2R_E}$$

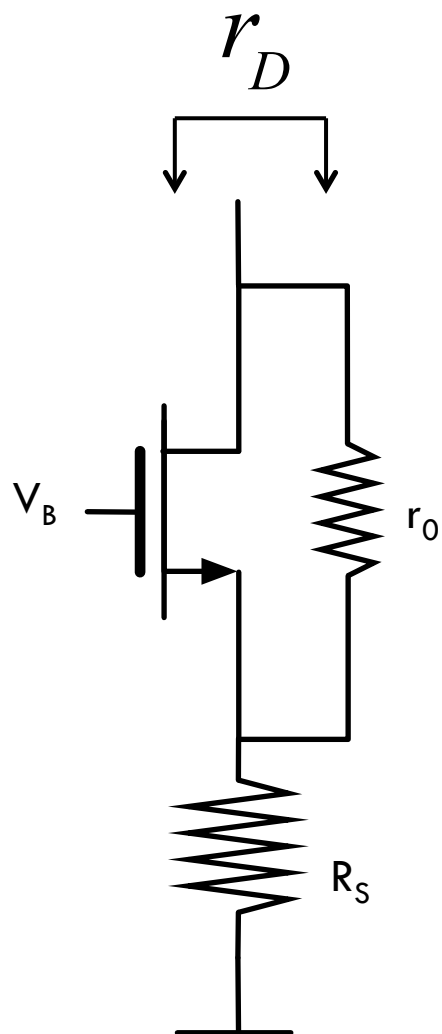
$$A_{cm} \approx -\frac{IR_L}{2IR_E} = -\frac{V_L}{V_{RE}} = -1$$



$$CMRR = \frac{V_a}{V_T}$$

Prestazioni limitate!

Generatori di corrente



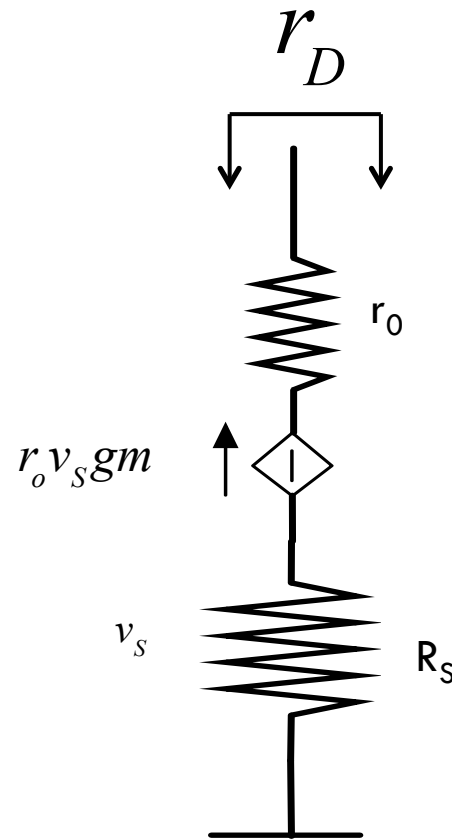
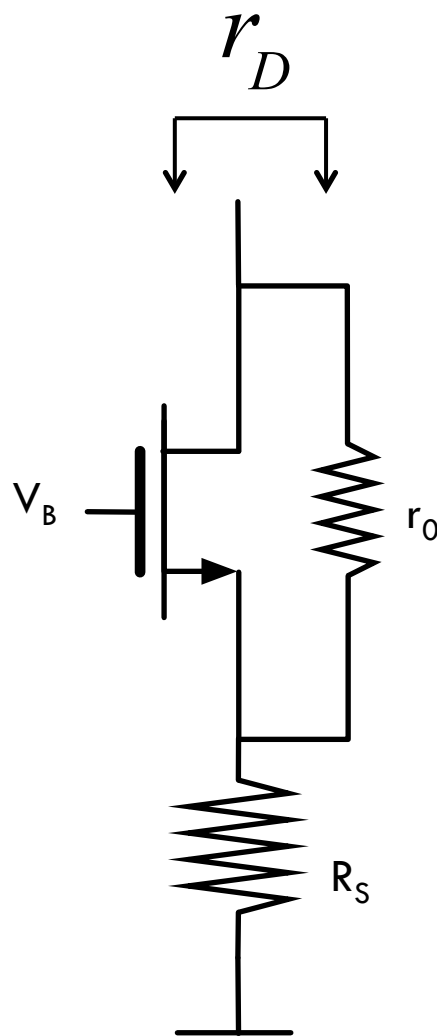
Positive loop approach

$$r_D = \frac{r_o + 1/gm // R_S}{1 - G_{loop}} = \frac{r_o + 1/gm // R_S}{1 - \frac{R_S}{1/gm + R_S}} = r_o(R_S gm + 1) + R_S$$

Open loop approach (forzo in tensione con VD)

$$i_D = \frac{v_D}{r_o + 1/gm // R_S} \left(1 - \frac{R_S}{1/gm + R_S} \right)$$

Generatori di corrente



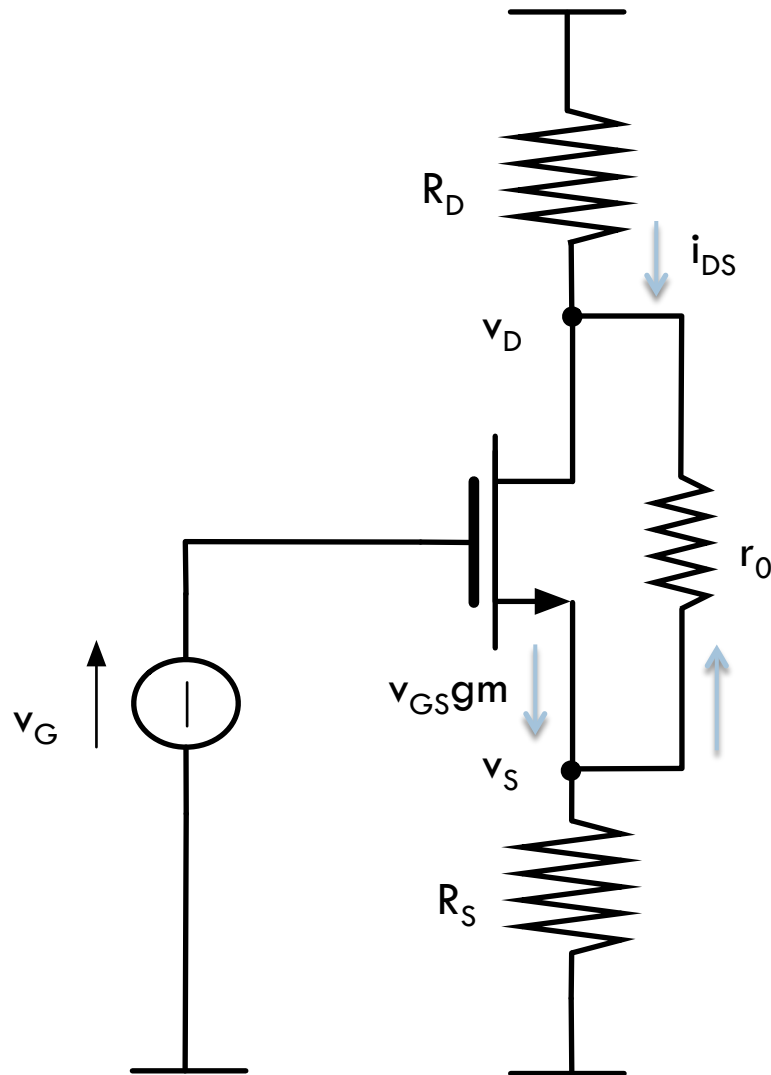
Approccio retroazione serie-serie

$$r_D = R_{OL}(1 - G_{loop})$$

$$= (R_S + r_o) \left(1 + \frac{g_m r_o R_S}{R_S + r_o} \right)$$

$$r_D = R_S + r_o (1 + g_m R_S)$$

Guadagno stadi elementari con r_o

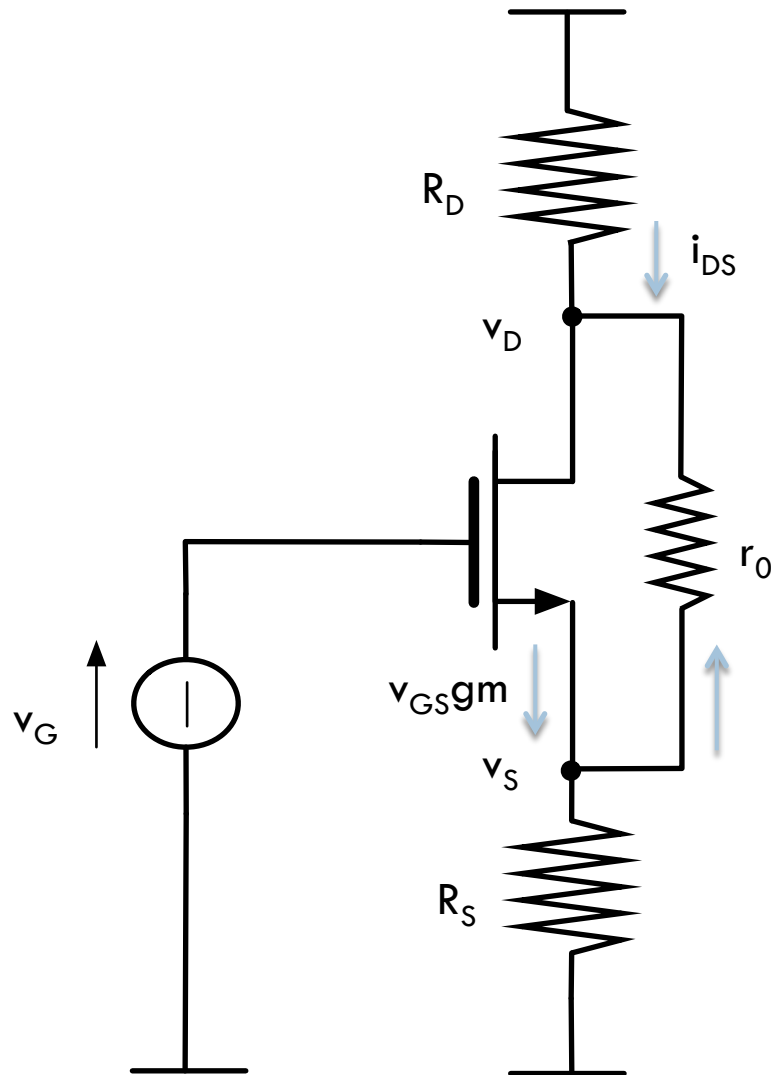


$$i_{DS} = v_{GS} gm + \frac{v_{DS}}{r_o}$$

$$i_{DS} = v_{GS} gm - i_{DS} \frac{R_S + R_D}{r_o}$$

$$i_{DS} = \frac{v_{GS}}{\frac{1}{gm} + \frac{R_S + R_D}{r_o gm}} = \frac{v_{GS}}{gm \left(1 + \frac{R_S + R_D}{r_o} \right)}$$

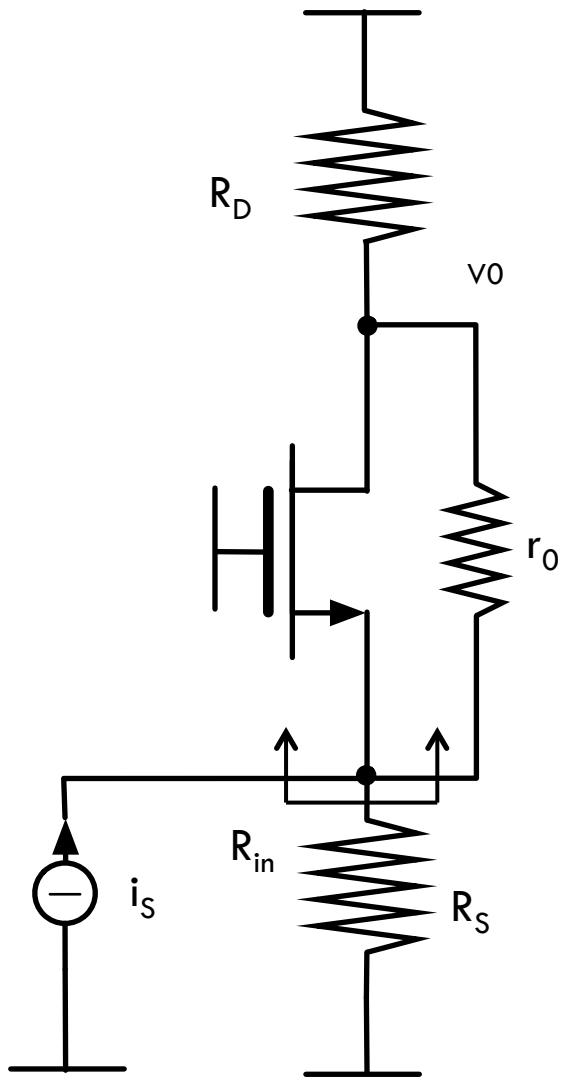
Guadagno stadi elementari con r_o



$$\frac{v_D}{v_G} = \frac{-R_D}{\frac{1}{gm} \left(1 + \frac{R_S + R_D}{r_o} \right) + R_S}$$

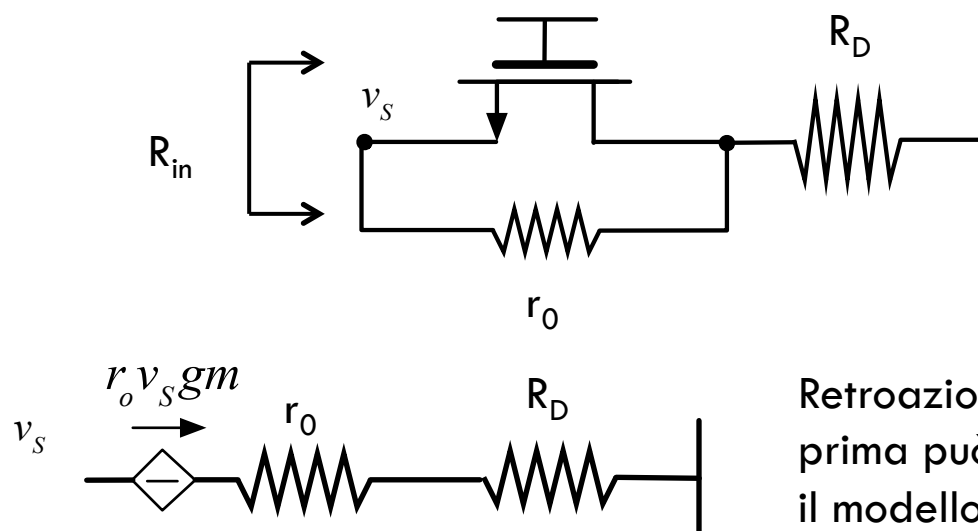
$$\frac{v_S}{v_G} = \frac{R_S}{\frac{1}{gm} \left(1 + \frac{R_S + R_D}{r_o} \right) + R_S}$$

Guadagno stadi elementari con r_o



$$v_o = i_s \frac{R_S}{R_S + R_{in}} R_D$$

Guadagno stadi elementari con r_o

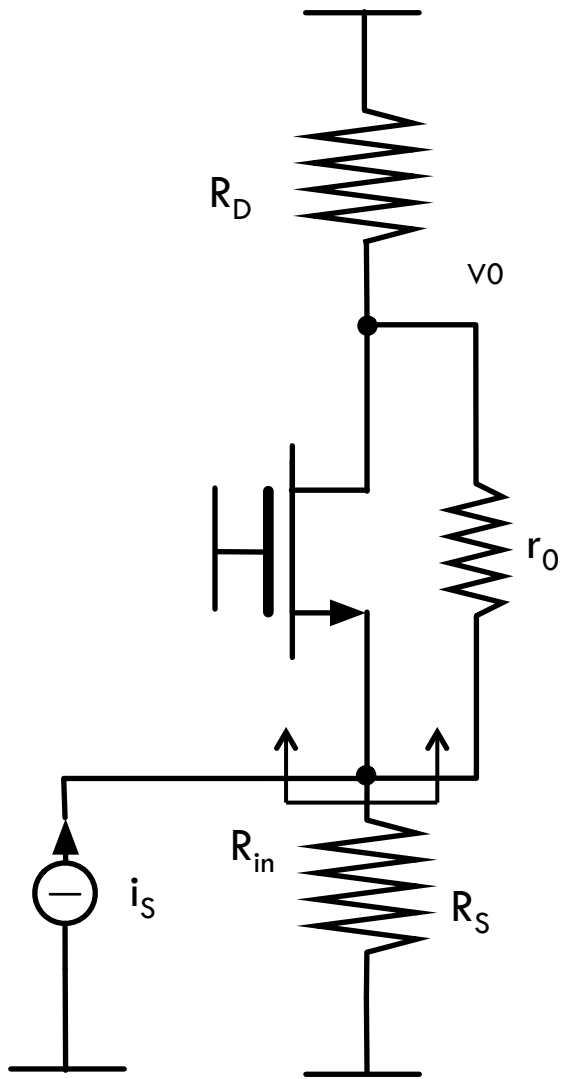


Retroazione negativa (analogamente a prima può essere interpretata utilizzando il modello con i generatori di corrente come retroazione positiva minore di 1)

$$R_{in} = \frac{r_o + R_D}{1 - G_{loop}(0)} = \frac{r_o + R_D}{1 + gmr_o}$$

Ovviamente tende a $1/gm$ se r_o è molto alta e maggiore di R_D .

Guadagno stadi elementari con r_o



$$v_o = i_s \frac{R_S}{R_S + R_{in}} R_D$$

$$v_o = i_s \frac{R_S}{R_S + \frac{r_o + R_D}{1 + g_m r_o}} R_D$$

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CURRENT MIRRORS



Current Mirrors

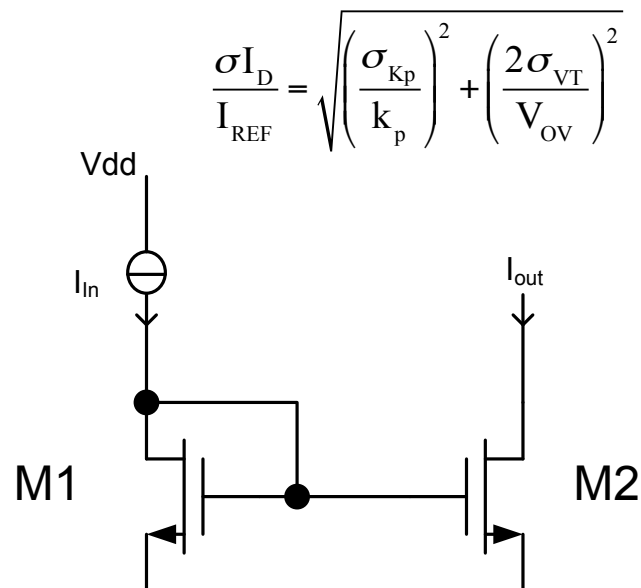
Small signal consideration

$$R_{in} = 1/gm_1$$

$$R_{out} = r_{o2}$$

$$V_{min} = V_{ov2}$$

Considering equal the two drain voltage

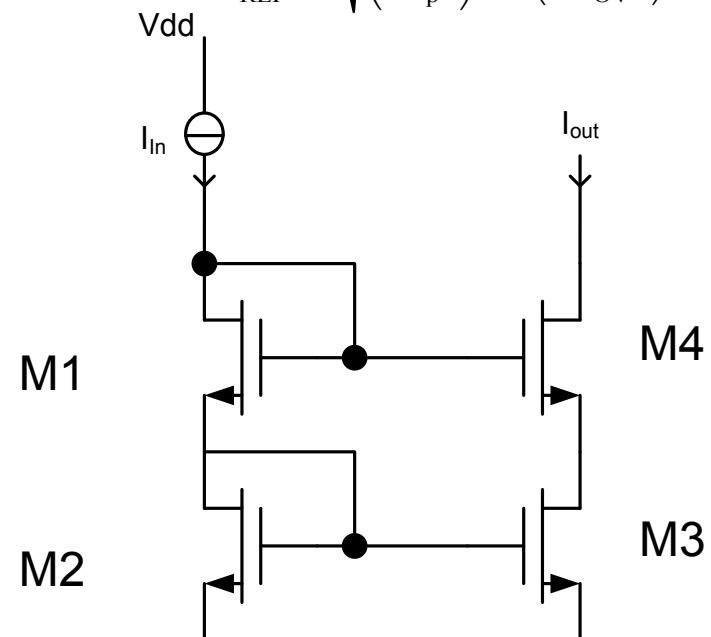


$$R_{in} = gm_1 gm_2 / (gm_1 + gm_2)$$

$$R_{out} = r_{o4} (1 + gm_4 r_{o3}) + r_{o3}$$

$$V_{min} = V_{gs3} + V_{ov4}$$

$$\frac{\sigma I_D}{I_{REF}} = \sqrt{\left(\frac{\sigma_{Kp}}{k_p}\right)^2 + \left(\frac{2\sigma_{VT}}{V_{OV}}\right)^2}$$



Current Mirror

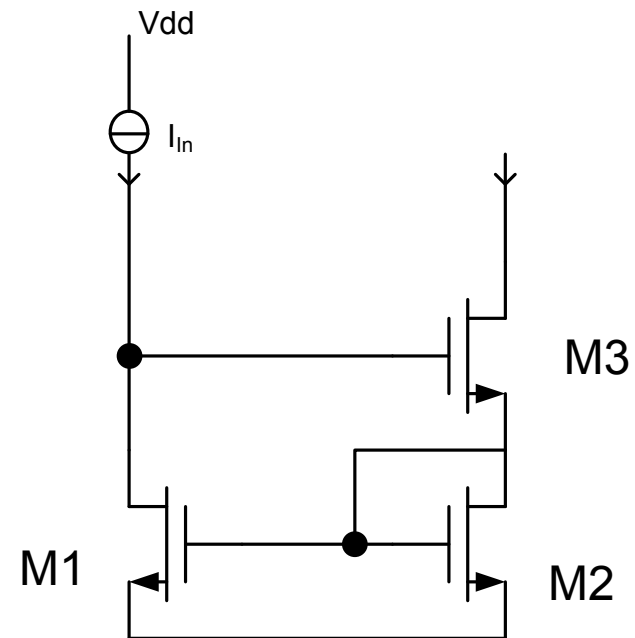
$$R_{in} \cong gm_2 gm_3 / (gm_3 + gm_2)$$

$$R_{out} \cong r_{o3} (1 + r_{o1} gm_3) \text{ with } (gm_1 \text{ equal to } gm_2)$$

$$V_{min} = V_{gs2} + V_{ov3}$$

Considering equal the two drain voltage

$$\frac{\sigma I_D}{I_{REF}} = \sqrt{\left(\frac{\sigma_{Kp}}{k_p}\right)^2 + \left(\frac{2\sigma_{VT}}{V_{OV}}\right)^2}$$



Current Mirror

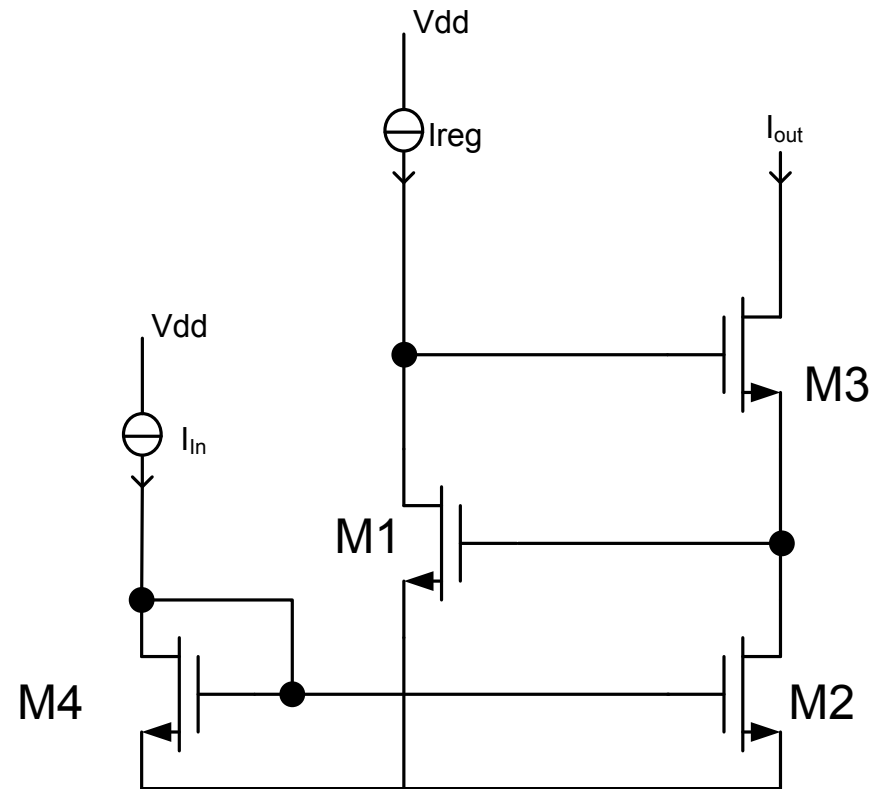
$$R_{in} = 1/gm_4$$

$$R_{out} \cong gm_3 r_{o3} r_{o2} gm_1 r_{o1}$$

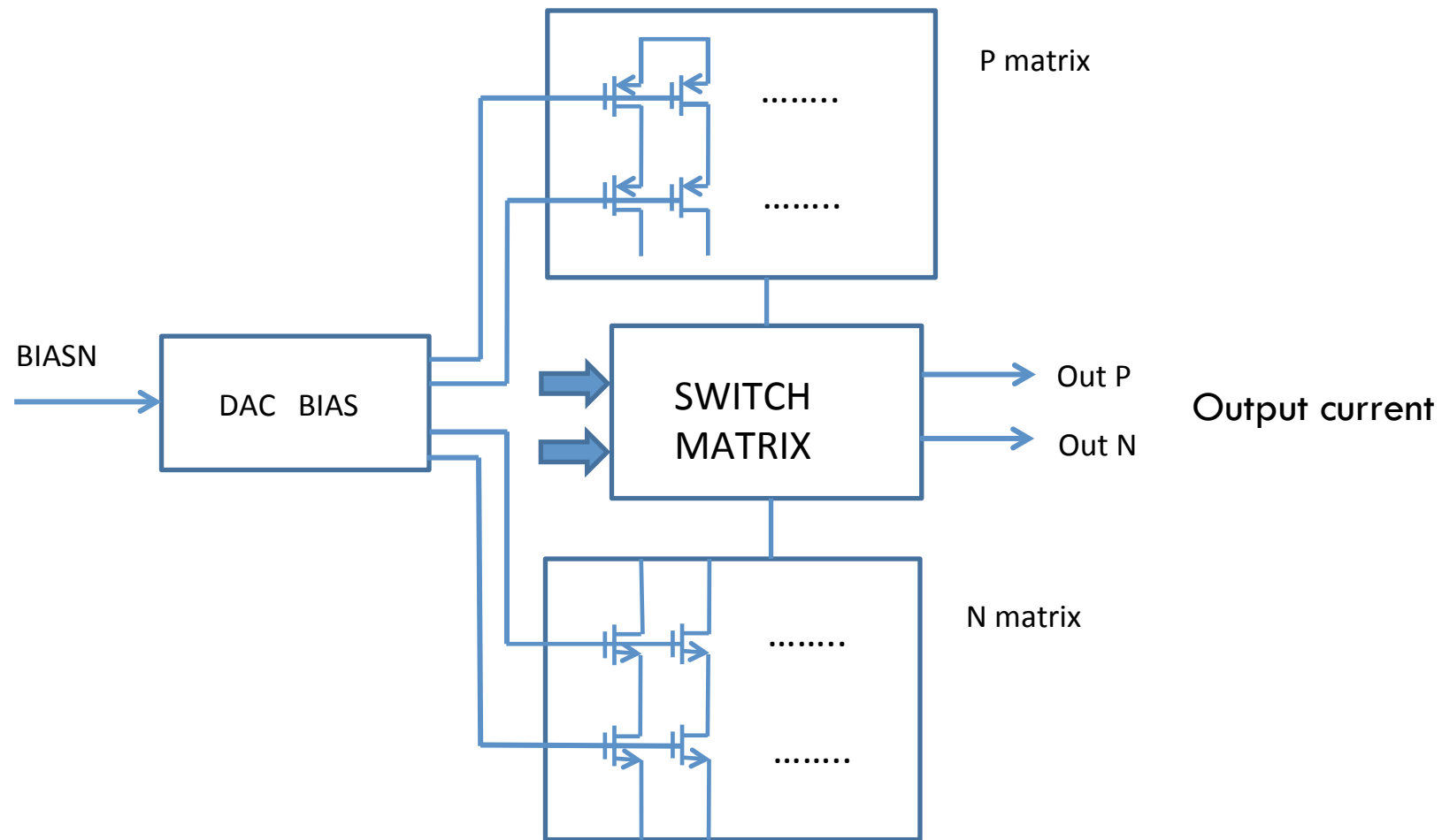
$$V_{min} = V_{ov3} + V_{gs1}$$

Considering equal the two drain voltage

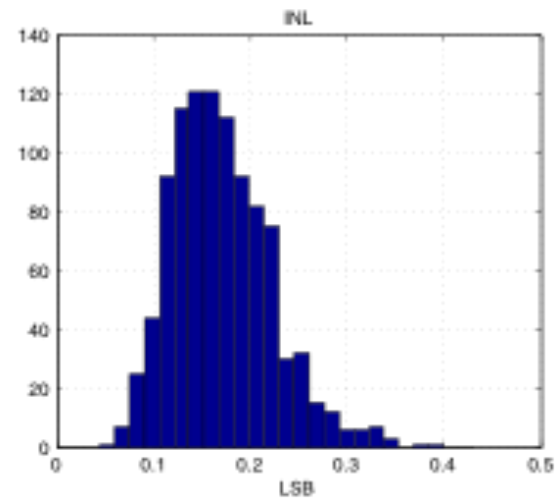
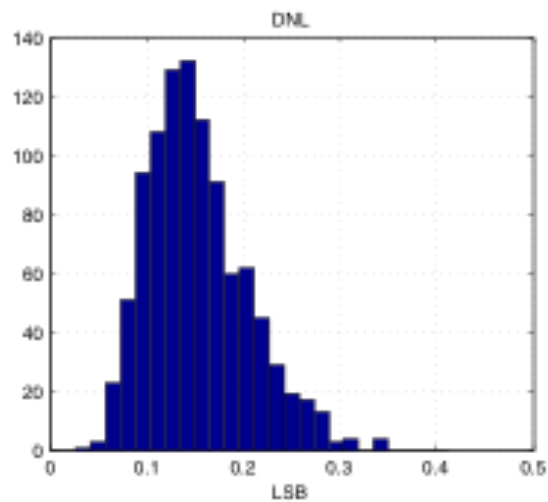
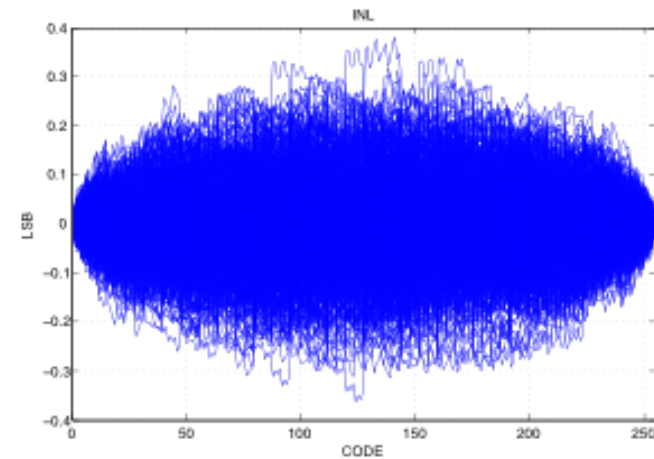
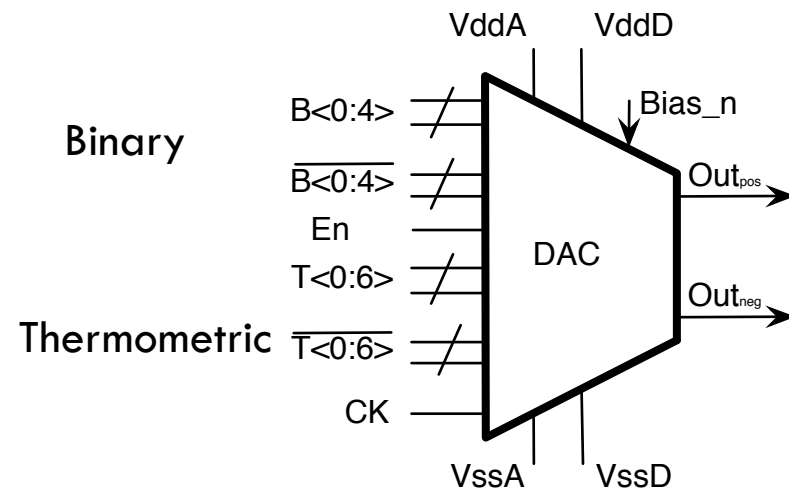
$$\frac{\sigma I_D}{I_{REF}} = \sqrt{\left(\frac{\sigma_{Kp}}{k_p}\right)^2 + \left(\frac{2\sigma_{VT}}{V_{OV}}\right)^2}$$



Current Steering DAC



Current Steering DAC



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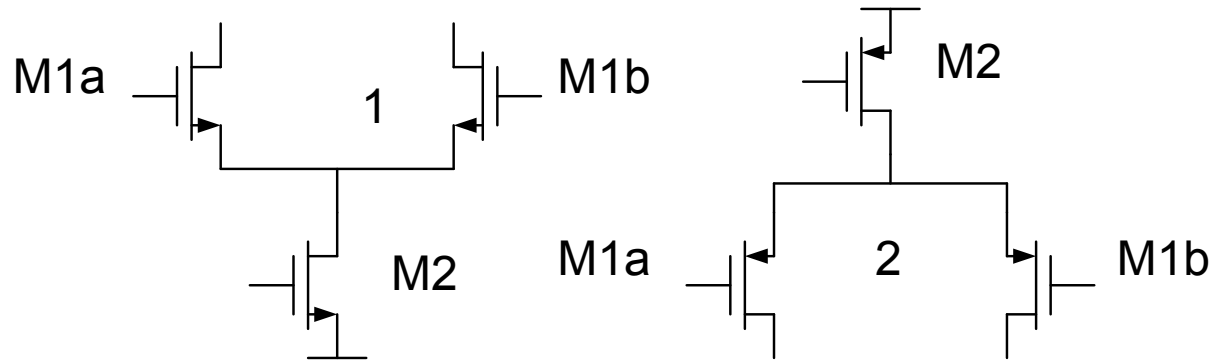
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SECOND STAGE

Assistant professor Stefano Saggini
Nome della conferenza o seminario

Differential stage



Input Dynamic

ICMR

1. $V_{incm} (min) = V_{ov1} + V_{ov2} + V_{Tn}$ (for n MOS) V_{ss} for (for p MOS)
2. $V_{incm} (max) = V_{dd} - V_{ov1} - V_{ov2} - |V_{Tp}|$ (for p MOS) V_{dd} for (for n MOS)

Transconductance :

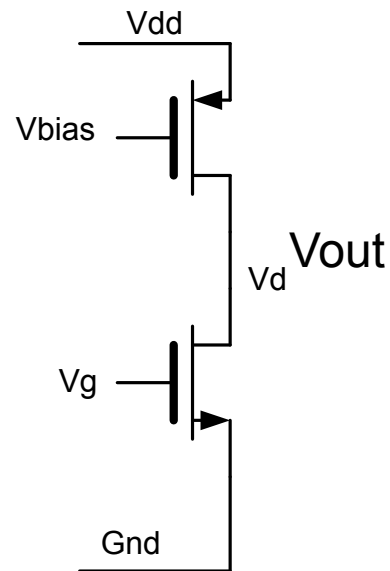
$$gm = \sqrt{K_1 I_{D2}}$$

Precision

$$\sigma_{Vos} = \sqrt{\sigma_{Vt1,2}^2 + \left(\frac{V_{OV1,2}}{2}\right)^2 \left[\left(\frac{\sigma_{Kp}}{k_p}\right)_{a,b}^2 + \left(\frac{\sigma_I}{I}\right)^2 \right]}$$

Specchio

Inverter current source



Output signal range

$$V_{out_{max}} = V_{dd} - V_{ovp}$$

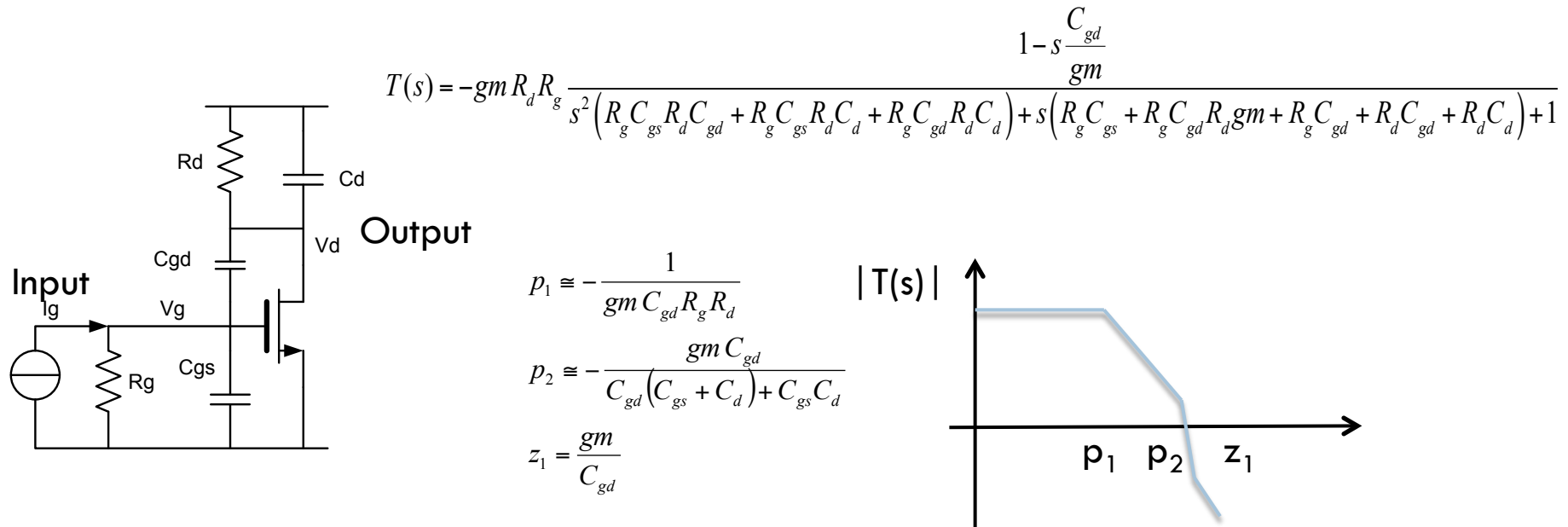
$$V_{out_{min}} = V_{ovn}$$

Input dynamics

$$V_{in_{min}} \cong V_{in_{max}} \cong V_{ovn} + VT$$

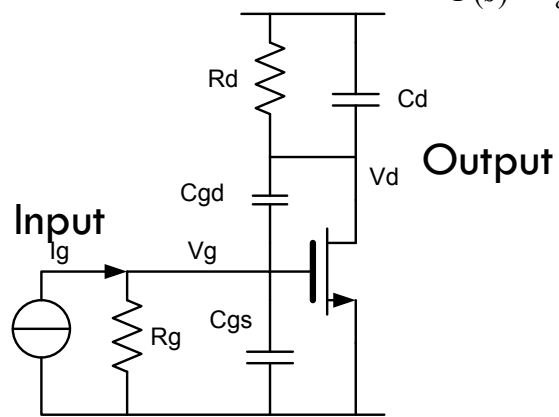
- To increase the dynamics
 - ▣ Increase W/L of nch and pch (decrease Vov)
 - ▣ Reduce the current

Inverter current source Tf



- The dominant pole p_1 can be calculated by using miller theorem
- To increase the performance
 - ▣ Increase W/L of nch increase gm and C_{gd}
 - ▣ Increase the current (increase of gm)

Pole splitting



$$T(s) = -gm R_d R_g \frac{1 - s \frac{C_{gd}}{gm}}{s^2 (R_g C_{gs} R_d C_{gd} + R_g C_{gs} R_d C_d + R_g C_{gd} R_d C_d) + s (R_g C_{gs} + R_g C_{gd} R_d gm + R_g C_{gd} + R_d C_{gd} + R_d C_d) + 1}$$

$$p_1 \cong -\frac{1}{gm C_{gd} R_g R_d}$$

$$p_2 \cong -\frac{gm C_{gd}}{C_{gd} (C_{gs} + C_d) + C_{gs} C_d}$$

$$z_1 = \frac{gm}{C_{gd}}$$

Adding a capacitor on Ccg

